

NEUTRONICS METHODS MEMO

Analytic ρR_D vs. OpenMC Transport — How They Fit Together

To: Kyle Keipper

From: Tom Mehlhorn (MECS)

Re: Neutron sampling from Helios → OpenMC, and where the Method-2 areal-density formula fits

Date: 2026-07-29 (updated 2026-07-31: $\sigma_{el,D}$ corrected to the ENDF/B-VIII.0 value)

Bottom line

Kyle's Method-2 areal density is the right deterministic ρR_D and we keep it. The adjustment is a change of role, not physics: Method 2 computes an analytic **diagnostic** (a scalar ρR_D), whereas OpenMC needs from Helios a **source** (birth positions + energy spectrum) plus **geometry** ($\rho(r)$ and composition). OpenMC then performs the same line integral by Monte Carlo and carries it beyond single scatter. Cleanest arrangement: use Method 2's ingredients to build the OpenMC source, let OpenMC transport, and keep Method-2 ρR_D as the independent benchmark.

What each side computes

Method 2 (analytic): for a neutron born at r_s along chord direction cosine μ , $r(l, \mu, r_s) = \sqrt{(r_s^2 + l^2 + 2 l r_s \mu)}$, and

$$\rho R_D(r_s, t) = (1/4\pi) \oint d\Omega \int_0^{l_{\text{exit}}(\mu)} \rho_D(r(l, \mu, r_s)) dl$$

($= \frac{1}{2} \int_{-1}^1 d\mu \int \rho_D dl$ for spherical symmetry), S -weighted over radius and burn-averaged in time. Exact for the mean deuterium column an isotropically emitted neutron traverses: no scattering, one species, frozen profile.

OpenMC: samples birth neutrons from $S(r, t)$ + fusion spectrum and transports through $\rho(r)$. It computes the same $\int \rho dl$ plus the full down-scatter spectrum, $(n, 2n)$, secondary photons, energy-angle correlations, and multiple scattering. Method 2's ingredients (sample $r_s \propto S$, isotropic emission) ARE the OpenMC source recipe; only the $\int \rho dl$ integral is made redundant by transport.

Formal correspondence + built-in validation

In the single-scatter, single-species, static-profile limit it is the same integral computed two ways. Expected D-elastic scatters per source neutron $C_D = (N_A/M_D) \cdot \sigma_{el,D} \cdot \rho R_D$, so

$$\rho R_D(\text{OpenMC}) = C_D \cdot M_D / (N_A \cdot \sigma_{el,D}), \quad \sigma_{el,D}(14.06 \text{ MeV}) \approx 0.641 \text{ b}$$

(ENDF/B-VIII.0 — the same value used in the findings note, not the ~ 0.8 b round number in earlier drafts). Run OpenMC on the same snapshot, tally D-elastic, convert; the two ρR_D should agree. Divergence (OpenMC a few % high at large ρR) is the multiply-scattered tail — the same residual slope in the first-principles $\rho R \leftrightarrow$ DSR coefficient, and the reason to run transport at high ρR . Agreement validates the source/geometry wiring; the gap measures the transport correction.

On that coefficient: with the DSR now taken on the traversed (Method-2) ρR , the self-consistent first-principles $\rho R \leftrightarrow$ DSR value on N210808 is 28.1 g/cm² per DSR (total, D+T+C) / 16.5 (fuel D+T only), bracketing the empirical NIF 20.4 — consistent with this being a clean over-igniting 1-D burn. See the findings note for the numbers and the reporting fix.

Hazard: do NOT pre-attenuate

Never use ρR_D to attenuate/scale the source before OpenMC — transport attenuates during the run, so a pre-attenuated source depletes the population twice. Same failure mode as the single-scatter double-count we

already fixed (undepleted primary + scattered $\rightarrow$ unphysical $>1.3\times$ escaping). Hand OpenMC the **undepleted** birth source (full yield, all channels); ρR_D stays a diagnostic check.

Assumptions to keep visible

1. **Volume Jacobian:** if S is per-volume emissivity, the radial birth pdf $p(r) \propto S(r) \cdot 4\pi r^2$. Method-2 weighting reads as per-shell ($\int S dr_s$) — keep conventions consistent or the source biases toward center.
2. **Frozen profile:** $\rho(r)$ at emission time. Excellent (neutron transit $\sim 1\text{--}2$ ps vs ~ 100 ps hydro) but an assumption.
3. **Burn-averaging is nonlinear:** ρR of the time-averaged profile $\neq$ time-average of per-snapshot ρR . Run a few emission-weighted snapshots, $S(t)$ -weight the tallies.
4. **Species:** Method 2 as drawn is D-only (right for fuel ρR_D). Down-scatter and the OpenMC geometry need all scatterers along the chord — D, T, and carbon.
5. **Isotropy:** fine for thermal DT.

Recommended workflow

One set of Helios inputs, two paths, cross-checked:

1. Per snapshot extract $r(\text{shells})$, $\rho(r)$, (w_D, w_T, w_C) , emissivity $S(r)$.
2. **Analytic (Kyle):** Method-2 ρR_D , S -weight over radius, burn-average $\rightarrow \rho R_D$.
3. **Transport (OpenMC):** nested-sphere geometry from $\rho(r)$ +composition; source from $S(r)$ (radial, isotropic) + fusion spectrum; feed the undepleted source; tally D-elastic, leakage spectrum, DSR.
4. **Compare:** convert OpenMC D-elastic $\rightarrow \rho R_D$ vs analytic. Agreement validates both; residual = multiple-scatter correction.
5. Hand the escaping source to the chamber OpenMC model (Kyle / Owen) for TBR / damage.

Implemented as a working sketch in [helios_to_openmc.py](#) (steps 3–4): builds geometry + $S(r)$ -weighted isotropic source, runs neutron transport, returns ρR_D from the D-elastic tally for a one-line comparison against Method 2.

What Helios hands OpenMC

- **Source spatial:** birth $\propto S(r, t)$ (with $4\pi r^2$ Jacobian), isotropic direction.
- **Source energy:** fusion birth spectrum — DT 14.06 (Muir/Ballabio) + DD 2.45 + TT continuum, exact per-channel yields, undepleted.
- **Geometry:** concentric shells with $\rho(r)$ and (w_D, w_T, w_C) at the chosen time(s).
- **NOT an input:** a pre-computed ρR . OpenMC produces it; Method 2 checks it.